Raspberry Pi 5 + Radxa Penta SATA Hat NAS Case

D DivideByAwesome

VIEW IN BROWSER

updated 4. 7. 2025 | published 4. 7. 2025

Summary

My take on a case for the Raspberry Pi 5 with the Radxa Penta SATA Hat. Includes build instructions and a parts list.

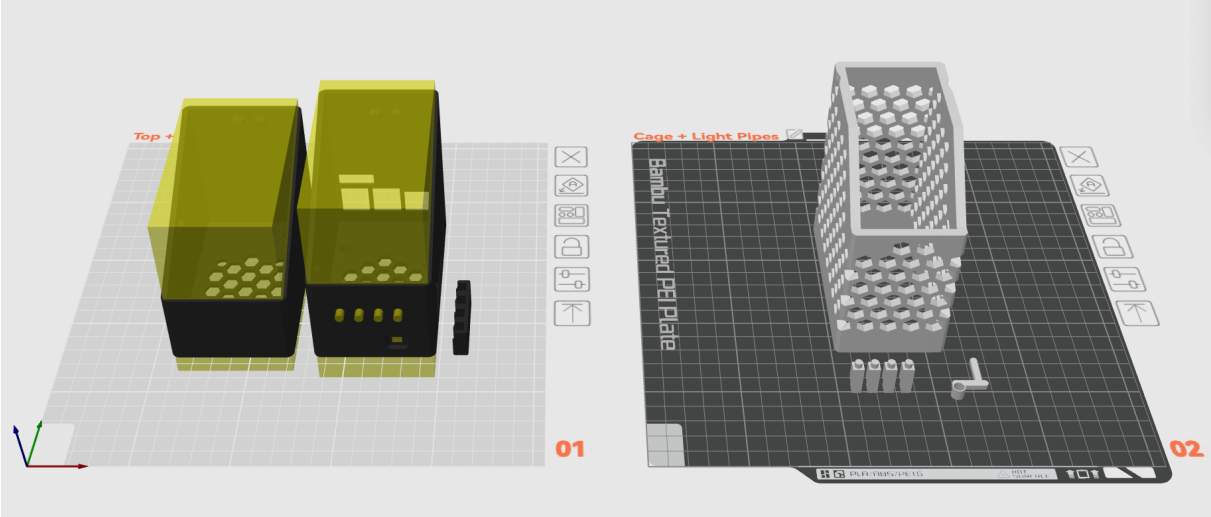
Gadgets > Computers

Tags: case pi raspberrypi raspberrypicase nas radxacase

Print Instructions

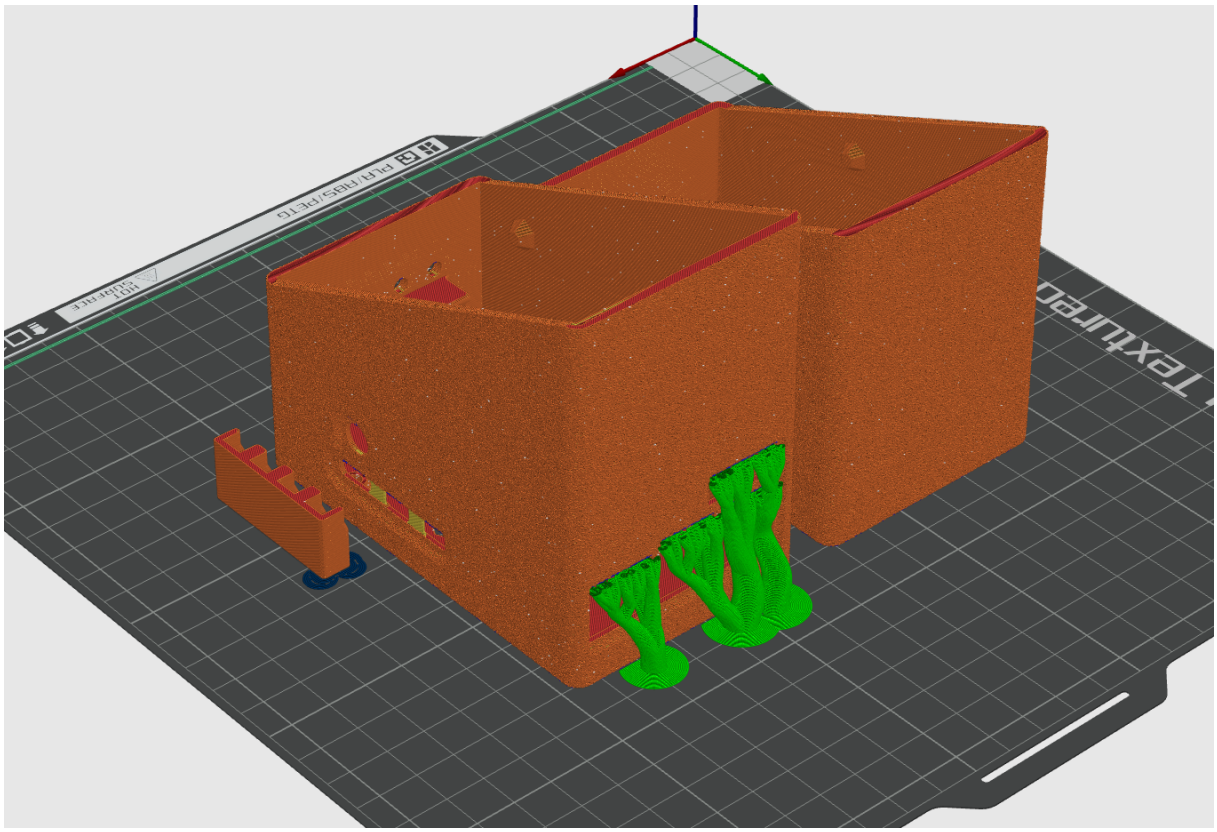
I recommend printing all components in PETG, ABS, ASA, or another more heat-tolerant plastic. The Pi can get pretty warm in the bottom of the case so printing in generic PLA could cause warping in some places.

The 3MF file has two plates. One is for the case top and bottom plus the light pipe cover. The other plate has the cage, light pipes, and power button. The parts are on separate plates since you'll probably want to print each plate in its own distinct color. I use black for plate 1 and transparent or white for plate 2.



For the top and bottom caps I added fuzzy skin to the exterior to improve the look. If you want to remove that, then you won't need the modifiers (those yellow boxes), which are there to remove the fuzzy skin from the inside of the caps. Fuzzy skin adds thickness to the walls and the internal tolerances are already pretty tight, so we don't want it on the interior. Also if you're printing on a non-textured build plate then you might want to remove the fuzzy skin to maintain a uniform look.

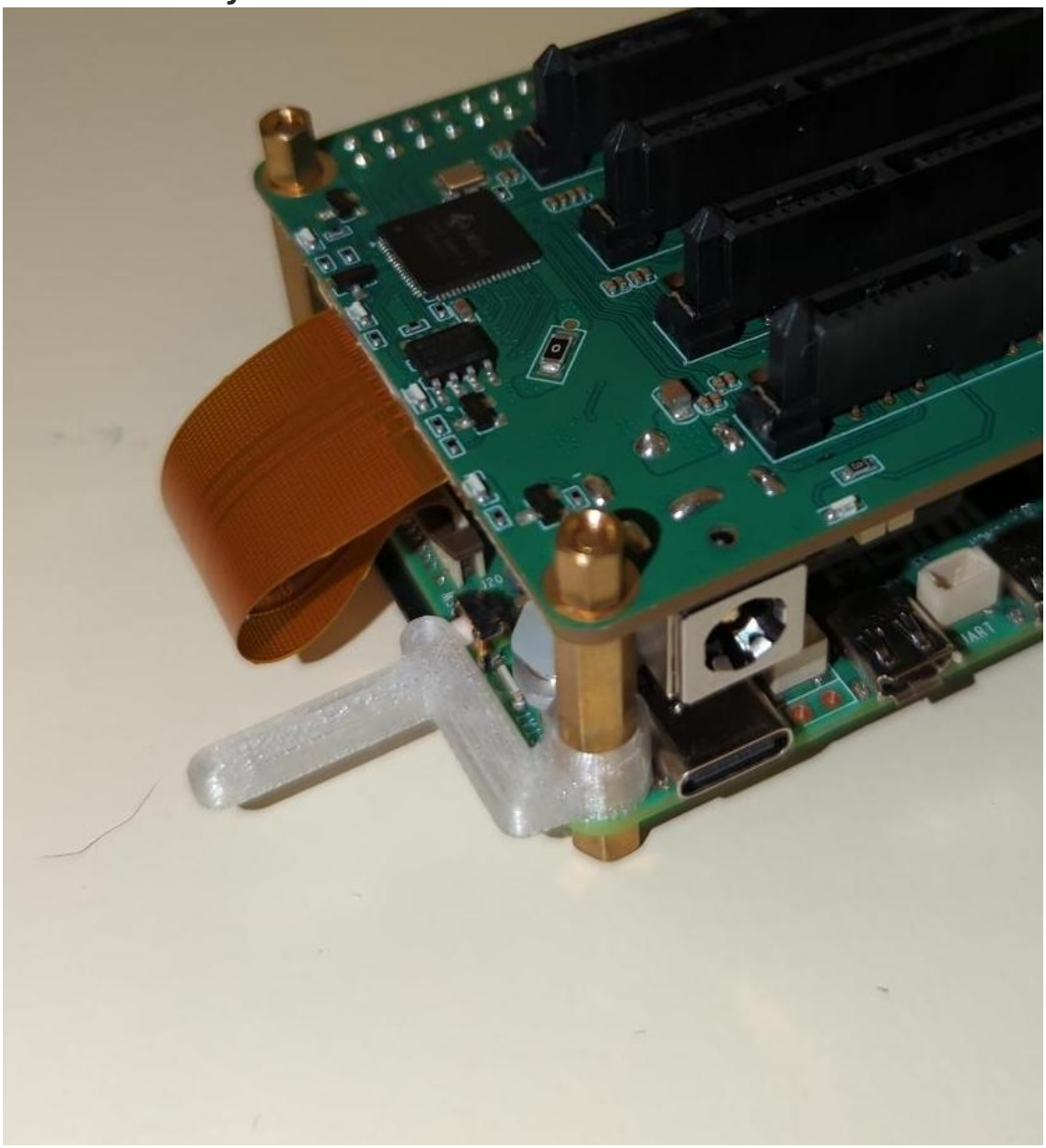
Leave the parts in the orientation defined by the 3MF file to minimize the need for supports. The necessary supports should already be painted on the backside of the bottom cap (where the IO is on the Pi). Besides that one spot, I have not had any issues printing without supports for the USB-C and other holes in the models but YMMV.



Build Instructions

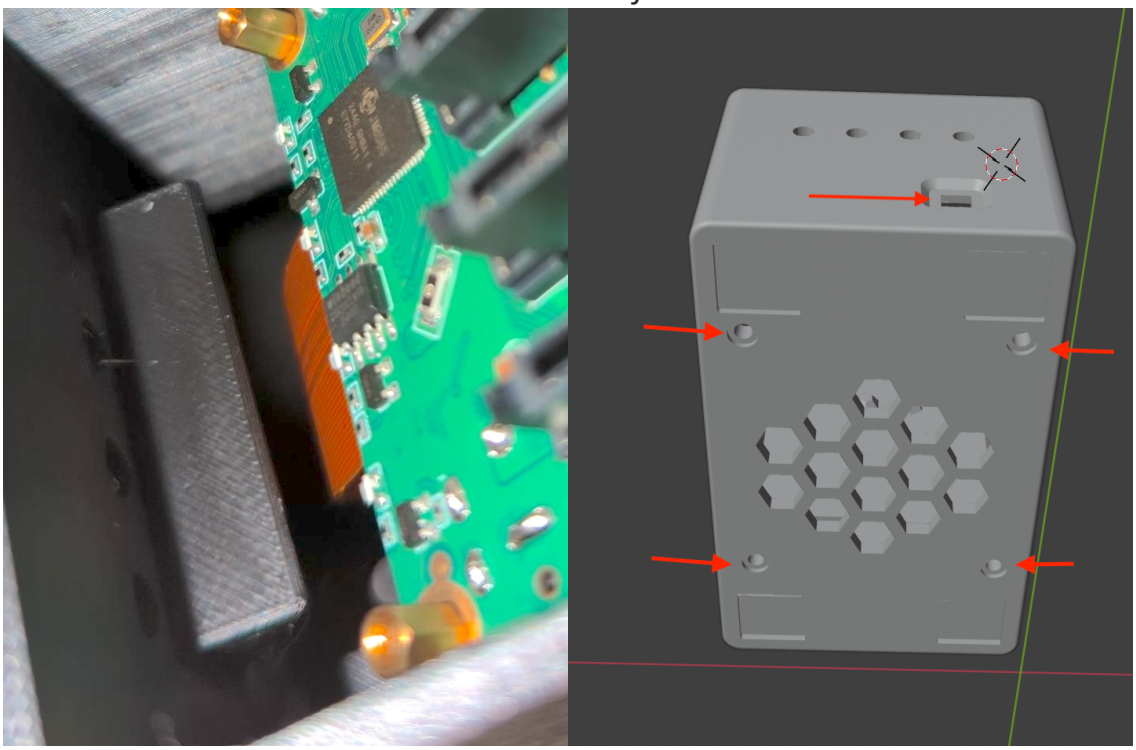
1. Connect the Raspberry Pi and Radxa Penta SATA Hat together. When connecting everything up make sure you slide the power button arm over the correct standoff as pictured. Also note that the shorter standoffs should be at the bottom of the assembly so that you can screw into them from underneath the case. **Do not install the**

drives or fan yet.

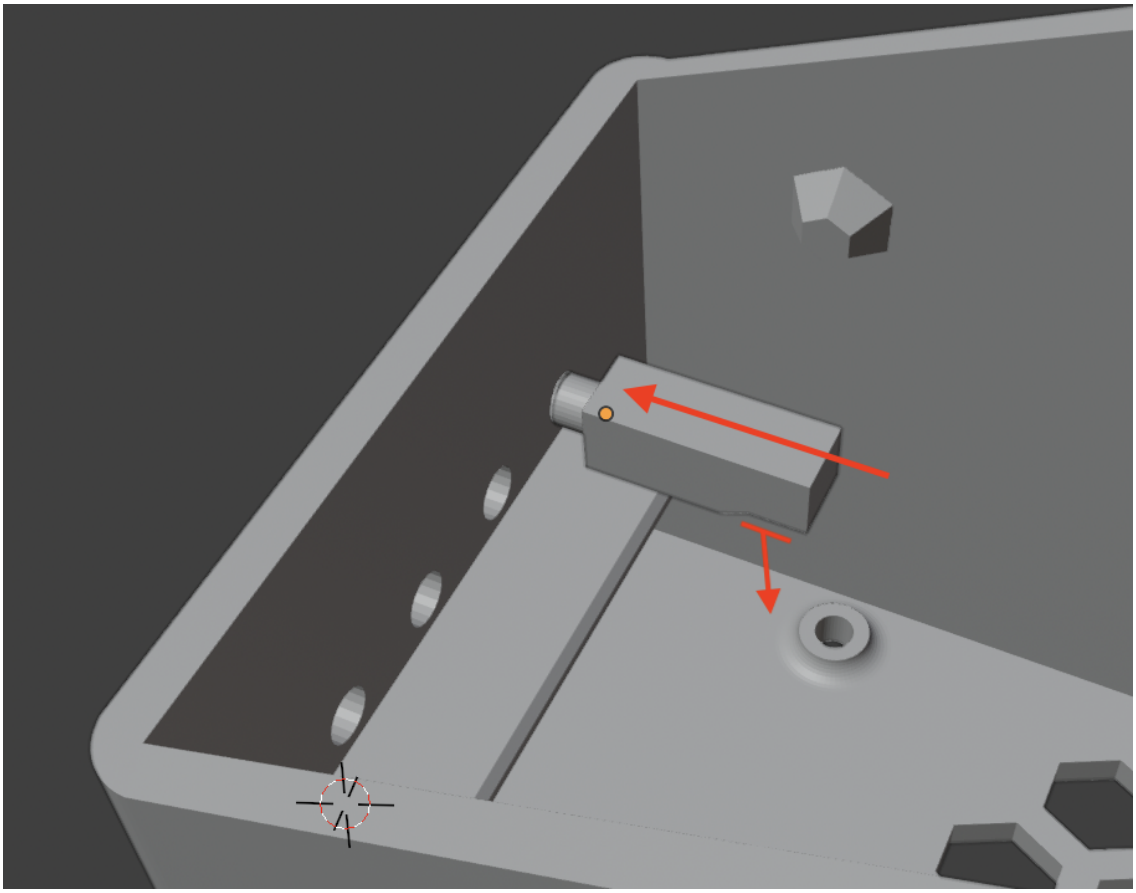


2. Add the 4 0.5in square bumpers to the underside of the bottom cap. There are square recesses on the bottom cap to allow for good alignment.
3. Carefully slide the assembly into the bottom cap. Rotate the Pi so that the ribbon cable fits under the angled shelf on the inside of the cap. The power button arm should slip out of the small rectangular opening. It might take a few attempts to get it right but it will fit. Just take your time and go slowly to prevent damaging the ribbon cable. Once the Pi is in position, secure it from underneath using the 4 M2.5 screws. Make sure the screws go through the cap and into the

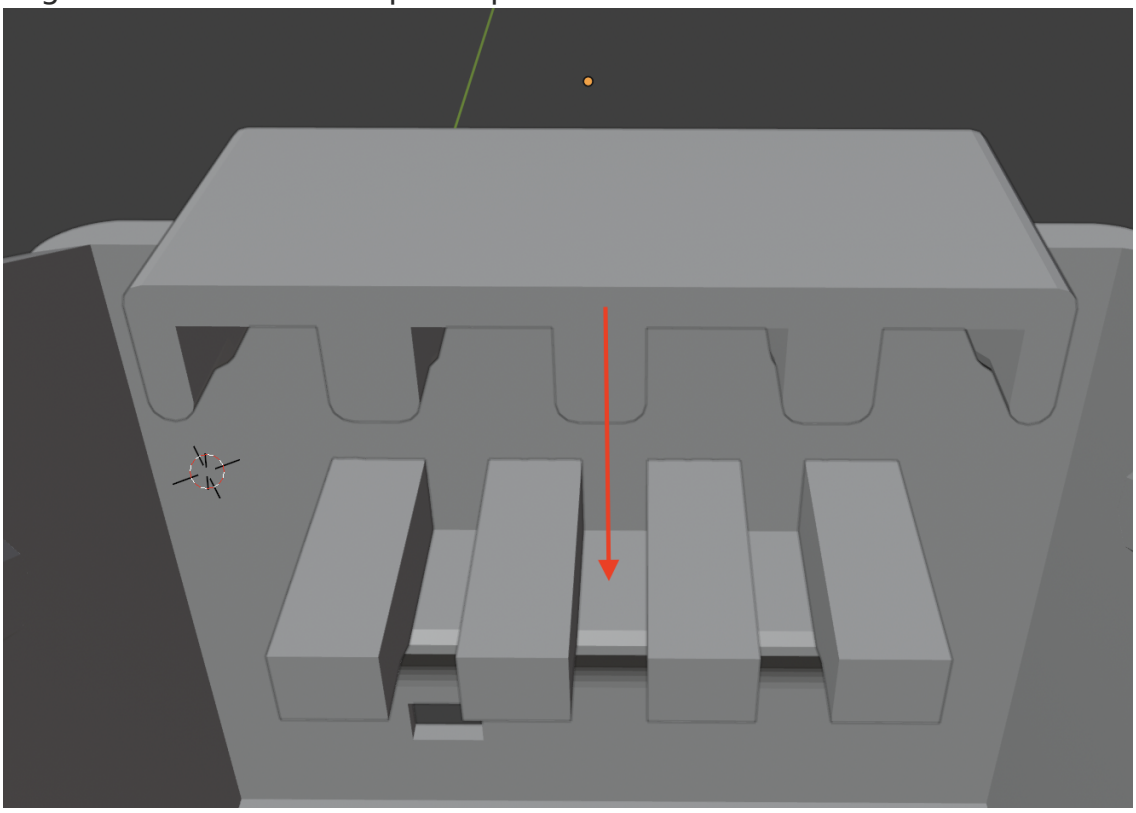
standoffs at the bottom of the assembly.



4. Install the light pipes on top of the angled shelf inside the bottom cap. The round end of the light pipe should fit into the holes in the cap. Make sure the round end of the light pipe is pushed all the way into the hole of the cap and that the notched end is facing down onto the Drive LEDs of the Hat.

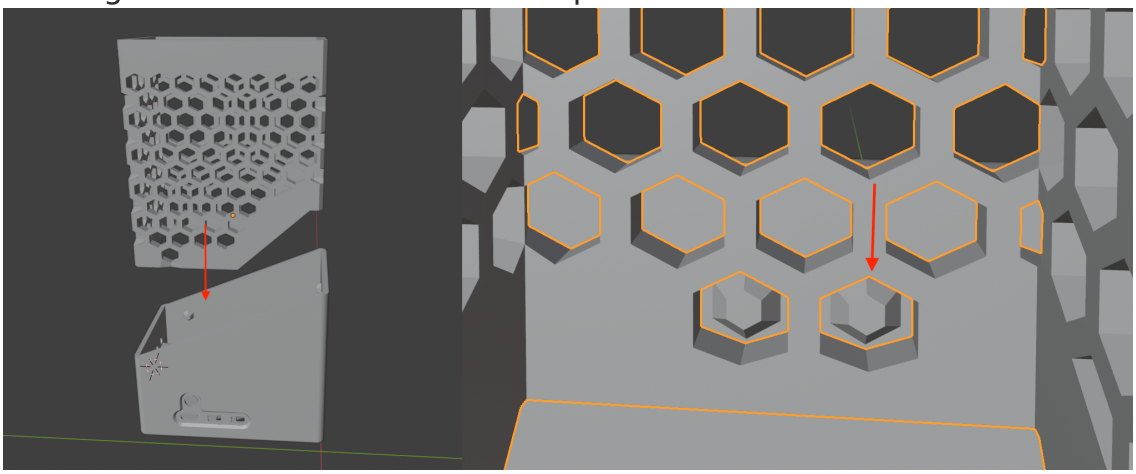


5. Position the light pipe cover over the light pipes making sure that it's flush up against the inner wall of the case. The teeth of the cover should fit between each light pipe. It's ok if it fits a little loosely. The cage and drives will keep it in position.



6. Install the Molex to 3pin fan converter into the Molex port on the Hat.
7. Connect the 40mm fan to the 3pin connector of the converter. Make sure there's plenty of cable to allow the fan to sit at the top of the case.
8. Install the 4 SATA drives into the SATA ports on the Hat.
9. (optional) Connect the 4 SATA drives together using the acrylic plate and 8 screws that came with the Radxa Penta SATA Hat kit. Make sure you only do this on the front of the drives as there will not be enough room in the back for the other acrylic plate and the Molex fan connector once the cage is in place.
10. Slide the cage over the drives and fan and push it down onto the bottom cap. Make sure it secures over the bumps in the interior of the bottom cap and that it's not pinching the fan cable and Molex converter. The angled end of the cage should face towards the IO side of the assembly. The solid parts of the cage will not be visible once

the cage is seated in the bottom cap.



11. Connect the fan to the underside of the top cap using 4 fan screws. I positioned my fan so it exhausts air out of the top but you might want to flip it so it pulls air into the case. Either orientation works. I have not tested to see which results in better cooling.
12. Push the top cap onto the cage. The bumps on the interior of the cap should align with the hexagons of the cage. Make sure the cap is seated all the way down on the cage. The gap between the top cap and bottom cap should be about 22-23mm. If the gap is over 25mm then you probably need to push the top cap down a little further or it might be pinching the fan cable somewhere.
13. Finally, connect up the barrel jack to power the Pi and enjoy your new NAS!

Parts

- Raspberry Pi 5 + MicroSD Card
- Raspberry Pi 5 Active Cooler
- Radxa Penta SATA Hat
- 4x 2.5in SATA Drives (I used 1 TB Samsung SSDs)
- 40mm 3pin fan (I used a noctua fan with the included fan screws)
- Molex to 3pin fan converter
- Power converter with a standard barrel jack
- 4x M2.5 screws
- 4x 0.5in Square bumpers
- 3d printed Top Cap
- 3d printed Bottom Cap
- 3d printed Hexagon Cage
- 4x 3d printed Light Pipes
- 3d printed Light Pipe Cover
- 3d printed Power Button Arm

Pi + Penta Hat build videos

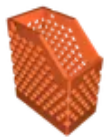
Here's a video of the build Jeff Geerling did of the pi and penta hat. I pretty much followed this when connecting and setting up the pi and penta hat. I'm also using OpenMediaVault for easy setup of connections. Just follow his video and once you have it all connected up then you should be able to add it to the case.

Here's another video by Michael Klements who did a build and a case for the pi + penta hat that I drew inspiration from (and used his power button).

Model files

v10.3mf

Exported from OrcaSlicer. This includes print settings and all models on two build plates



v10-cage.stl

v10-bottom_cap.stl

v10-light_pipe_cover.stl

v10-power_button_arm.stl

v10-light_pipe.stl

v10-top_cap.stl

License ©

This work is licensed under a
Creative Commons (4.0 International License)

Attribution-NonCommercial

✖ | Sharing without ATTRIBUTION
✔ | Remix Culture allowed
✖ | Commercial Use
✖ | Free Cultural Works
✖ | Meets Open Definition